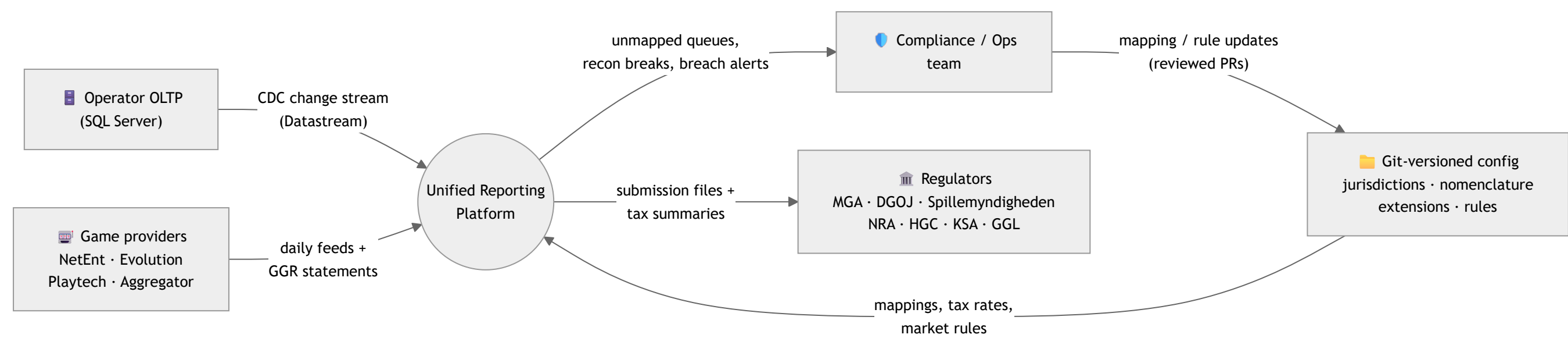


Data Flow Diagram — Unified Regulatory Reporting Platform

How data moves from the operator's transactional systems, through the layered transformation DAG, past the compliance gate, and out to each regulator — with **config as data** driving the per-market fan-out. Three views:

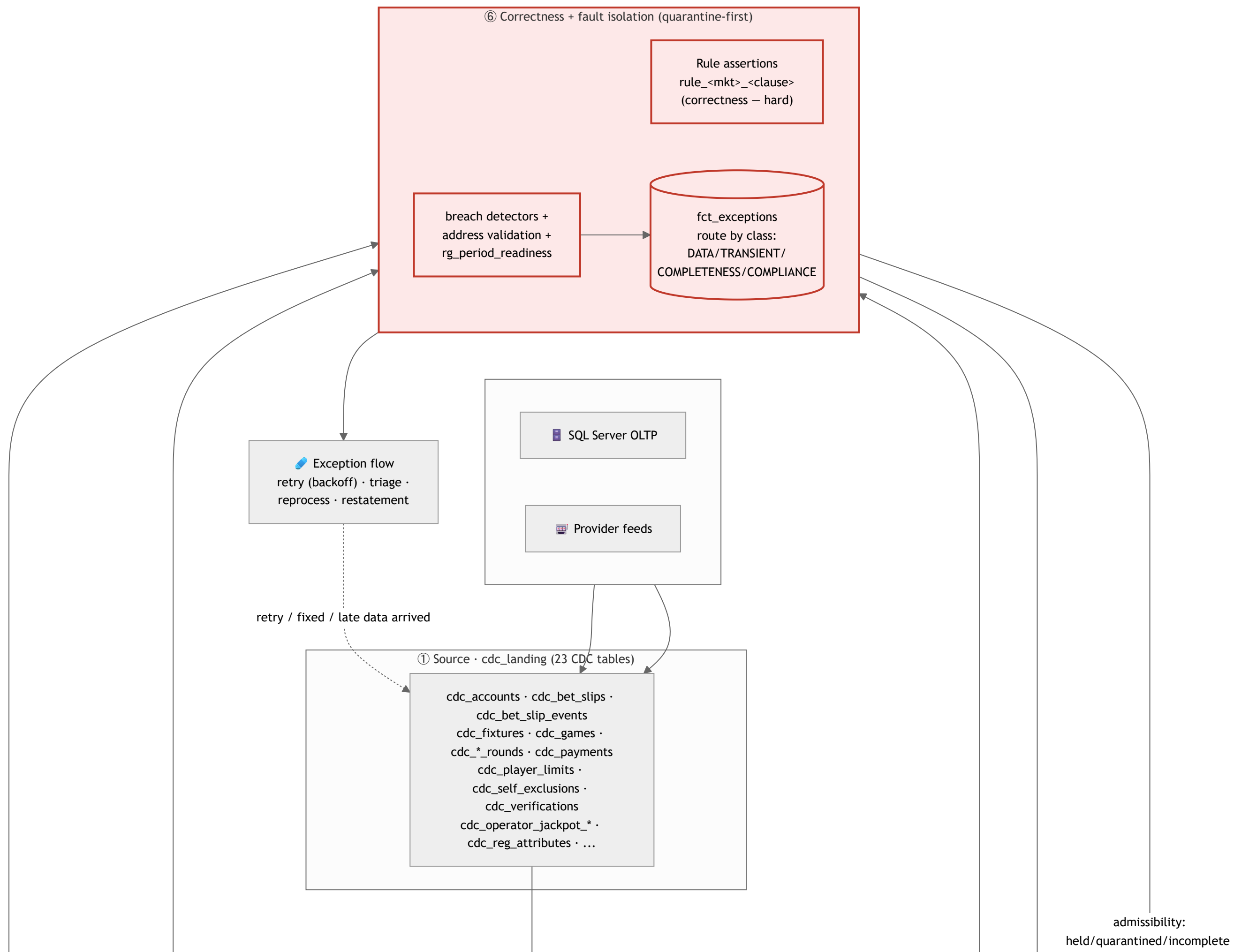
- 1. **Context (Level 0)** — external entities and the system boundary
- 2. **Pipeline (Level 1)** — the layered processes, data stores, and the gate
- 3. **Config control plane** — how "variance as data" parameterises the DAG

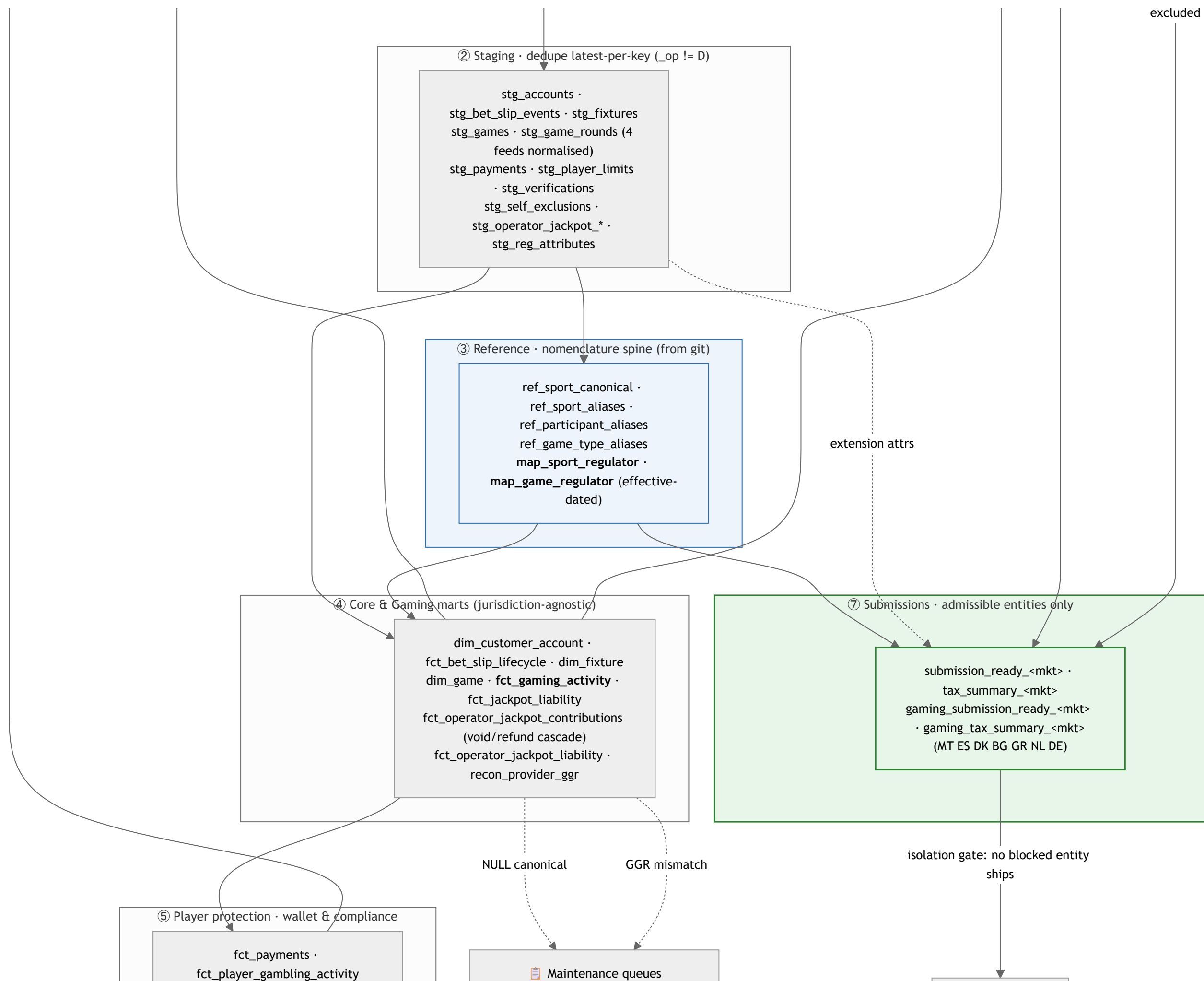
1 • Context diagram (Level 0)

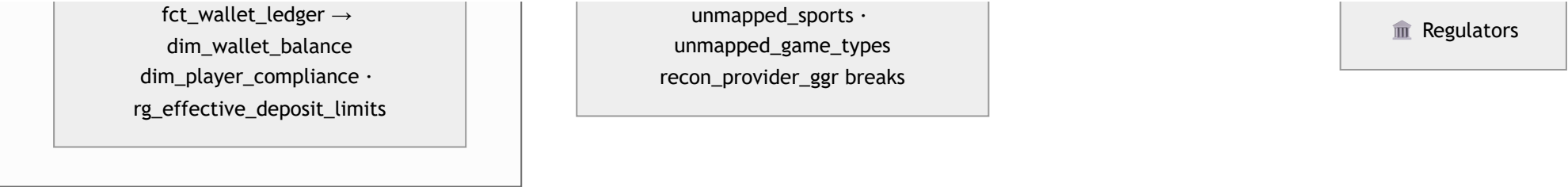


2 • Pipeline (Level 1) — the transformation DAG

Data flows top-to-bottom through materialised layers. Between the marts and the regulator outputs sit **correctness rules** (hard-fail on invalid data) and **fault isolation** (quarantine-first): a bad/late/held entity is routed to `fct_exceptions` and excluded, while everyone else ships. **Nothing held/quarantined/incomplete ever reaches a regulator** — the one hard block.



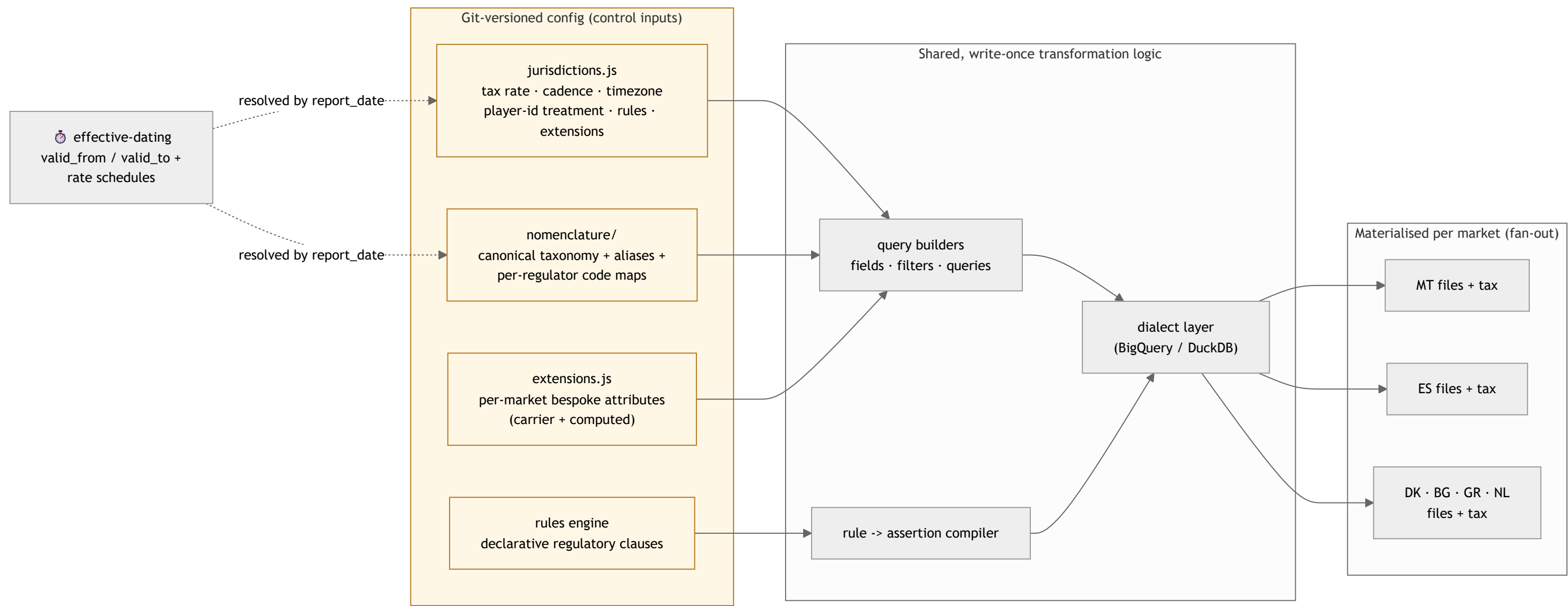




Reading the gate (quarantine-first): correctness rules still hard-fail on invalid data. But a breach, a bad postcode, or an unready period no longer aborts the run — the entity is routed into `fct_exceptions` (DATA→quarantine, TRANSIENT→retry, COMPLETENESS→wait, COMPLIANCE→hold) and excluded from its file by the admissibility filter, while everyone else ships. The one hard structural block is **isolation itself**: no held/quarantined/incomplete entity may reach a regulator. Legitimately-absent data (an OPEN slip's settlement) is shipped correctly as empty — told apart from "late" by terminal *state*, not row-absence. **Negative tests** prove each guardrail — including the isolation gate — fires.

3 · Config control plane — "market variance is data"

The same DAG serves every market. What differs between MT, ES, DK, BG, GR and NL is **configuration**, not code — so a new market is a config change, not a new pipeline.



Effective-dating closes the time dimension: a resubmission of a historical period resolves the tax rate and regulator code that were *in force then* (`valid_from` / `valid_to` on the maps, a rate schedule on the jurisdiction), not today's values.

4 · Runtime paths — production vs offline

The identical model runs two ways; the offline path is the developer gate and needs no cloud.

